

T3 BSM UV and matched EFT contributions

EFT order: 5; loop order: 1. The EFT expressions show $\Delta\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{matched}}^{\text{SM+BSM}} - \mathcal{L}_{\text{matched}}^{\text{SM}}$.

$$T3 - B, \quad \alpha = -1 : \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (1, 0)$$

$$\begin{aligned}
& \text{BSM UV Lagrangian} - \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_{S_1}^2 \\
& + \frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1}) \\
& - \frac{1}{2} N_{r_1}^T C N_{r_1} M_F \\
& + \overline{\ell_{a_1, i_1}} P_R N_{r_1} \text{NewScalar1}_{a_1} y_{1, i_1, r_1} \\
& + \overline{y_{2, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \text{NewScalar2}_{a_2} \bar{e}_{a_1, a_2} \\
& + \overline{y_{1, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \text{NewScalar1}_{a_1}^\dagger \\
& + \text{lambdaH1Inv1} H_{a_3} \text{NewScalar1}_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \text{lambdaH1Inv2} H_{a_3} \text{NewScalar1}_{a_4} \text{T3HiggsPortal1CGInv2}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{a_3} \text{NewScalar1}_{a_4} \text{T3Scalar1SelfCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \overline{\lambda_{T3}} \text{NewScalar2}_{a_1} \epsilon_{a_2, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{a_3}^\dagger \\
& - M_{S_2}^2 \text{NewScalar2}_{a_1} \text{NewScalar2}_{a_1}^\dagger \\
& + \overline{\ell_{a_2, i_1}} P_R N_{r_1} y_{2, i_1, r_1} \epsilon_{a_2, a_1} \text{NewScalar2}_{a_1}^\dagger \\
& + H_{a_1} H_{a_2} \text{NewScalar1}_{a_3} \lambda_{T3} \bar{e}_{a_2, a_3} \text{NewScalar2}_{a_1}^\dagger \\
& + \text{lambdaH2Inv1} H_{a_3} \text{NewScalar2}_{a_4} \text{T3HiggsPortal2CGInv1}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambdaH2Inv2} H_{a_3} \text{NewScalar2}_{a_4} \text{T3HiggsPortal2CGInv2}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambda12Inv1} \text{NewScalar1}_{a_3} \text{NewScalar2}_{a_4} \text{T3CrossScalarCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambda12Inv2} \text{NewScalar1}_{a_3} \text{NewScalar2}_{a_4} \text{T3CrossScalarCGInv2}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{a_3} \text{NewScalar2}_{a_4} \text{T3Scalar2SelfCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar2}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar1}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{a_1}) \\
& + D_{\mu_1} (\text{NewScalar2}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{a_1}) \\
& \Delta \mathcal{L}_{\text{EFT}} - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{2, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1}} \ell_{a_3, i_2}^T C P_L \ell_{a_2, i_1} H_{a_1} H_{a_4} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1, a_2} \bar{e}_{a_3, a_4} M_F^3}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1}} \ell_{a_3, i_2}^T C P_L \ell_{a_2, i_1} H_{a_1} H_{a_4} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1, a_2} \bar{e}_{a_3, a_4} M_F^3}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{T3} \ell_{a3, i_1}} C P_R \overline{\ell_{a4, i_2}}^T M_{S2}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_4, a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{T3} \ell_{a3, i_1}} C P_R \overline{\ell_{a4, i_2}}^T M_{S1}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_4, a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{4 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{2 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{2 (M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{2 (M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{4 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^2}{2 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{\hbar \overline{y_{1, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \overline{y_{1, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} M_{S1}^2 y_{1, i_3, r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} M_{S1}^2 y_{1, i_3, r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} y_{2, i_3, r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} y_{2, i_3, r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \overline{y_{2, i_1, r_1} Y_{e, i_3, i_2} e_{i_2}} P_L \ell_{a_1, i_1} M_{S2}^2 y_{2, i_3, r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S2}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{\hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{12 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4 \epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_1}^2 \right)} \\
& - \frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_1}^2 \right)} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_1}^2 \right)^2} \\
& - \frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_1}^2 \right)^2} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \epsilon}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{8}\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S_2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_2}^2\right)} \\
& -\frac{\hbar\log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right)\overline{y_{2,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S_2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_2}^2\right)} \\
& -\frac{\overline{hy_{2,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S_2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\left(M_F^2-M_{S_2}^2\right)^2} \\
& -\frac{\hbar\log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right)\overline{y_{2,i_3,r_1}\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_{S_2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\left(M_F^2-M_{S_2}^2\right)^2} \\
& +C_{H2}H_{a_1}H_{a_1}^\dagger \\
& +\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8}\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S_1}^2\right)} \\
& -\frac{\hbar\log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S_1}^2\right)} \\
& -\frac{\overline{hy_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{4\left(M_F^2-M_{S_1}^2\right)^2} \\
& -\frac{\hbar\log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{4\left(M_F^2-M_{S_1}^2\right)^2} \\
& +\frac{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8}\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_2}^2y_{2,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S_2}^2\right)} \\
& -\frac{\hbar\log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_2}^2y_{2,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S_2}^2\right)} \\
& -\frac{\overline{hy_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_2}^4y_{2,i_3,r_1}H_{a_1}^\dagger}{4\left(M_F^2-M_{S_2}^2\right)^2} \\
& -\frac{\hbar\log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S_2}^4y_{2,i_3,r_1}H_{a_1}^\dagger}{4\left(M_F^2-M_{S_2}^2\right)^2} \\
& -H_{a_1}\mu^2H_{a_1}^\dagger \\
& +\frac{\text{lambdaH1Inv}l^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}
\end{aligned}$$

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$$\begin{aligned}
& + \frac{5\text{lambdaH1Inv}2^2\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& + \frac{\text{lambdaH2Inv}1^2\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{5\text{lambdaH2Inv}2^2\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& + \frac{\text{lambdaH1Inv}1\text{lambdaH1Inv}2\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{\text{lambdaH2Inv}1\text{lambdaH2Inv}2\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{1}{2}\text{lambdaH1Inv}1^2\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH1Inv}2^2\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv}1\text{lambdaH1Inv}2\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{1}{2}\text{lambdaH2Inv}1^2\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH2Inv}2^2\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv}1\text{lambdaH2Inv}2\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{\hbar\overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{\hbar\overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{\hbar\overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2}
\end{aligned}$$

$$T3 - C, \quad \alpha = -1 : \quad S_1 = (2, -\tfrac{1}{2}), \quad S_2 = (2, \tfrac{1}{2}), \quad F = (3, 0)$$

$$\text{BSM UV Lagrangian} - \text{NewScalar}_{1a_1} \text{NewScalar}_{1a_1}^\dagger M_{S1}^2$$

$$+ \frac{1}{2} i N_{A_1, r_1}{}^T C \gamma^{\mu_1} D_{\mu_1} (N_{A_1, r_1})$$

$$- \frac{1}{2} N_{A_1, r_1}{}^T C N_{A_1, r_1} M_F$$

$$\begin{aligned}
& + \overline{\ell_{a_1, i_1}} P_R N_{A_1, r_1} \text{NewScalar1}_{a_2} y_{1, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \\
& + \overline{y_{2, i_1, r_1}} N_{A_1, r_1} {}^T C P_L \ell_{a_3, i_1} \text{NewScalar2}_{a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3} \bar{\epsilon}_{a_1, a_2} \\
& + \overline{y_{1, i_1, r_1}} N_{A_1, r_1} {}^T C P_L \ell_{a_2, i_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \text{NewScalar1}_{a_1}^\dagger \\
& + \text{lambdaH1Inv1} H_{a_3} \text{NewScalar1}_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \text{lambdaH1Inv2} H_{a_3} \text{NewScalar1}_{a_4} \text{T3HiggsPortal1CGInv2}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{a_3} \text{NewScalar1}_{a_4} \text{T3Scalar1SelfCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \overline{\lambda_{\text{T3}}} \text{NewScalar2}_{a_1} \epsilon_{a_2, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{a_3}^\dagger \\
& - M_{\text{S2}}^2 \text{NewScalar2}_{a_1} \text{NewScalar2}_{a_1}^\dagger \\
& + \overline{\ell_{a_3, i_1}} P_R N_{A_1, r_1} y_{2, i_1, r_1} \epsilon_{a_2, a_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_3, a_2} \text{NewScalar2}_{a_1}^\dagger \\
& + H_{a_1} H_{a_2} \text{NewScalar1}_{a_3} \lambda_{\text{T3}} \bar{\epsilon}_{a_2, a_3} \text{NewScalar2}_{a_1}^\dagger \\
& + \text{lambdaH2Inv1} H_{a_3} \text{NewScalar2}_{a_4} \text{T3HiggsPortal2CGInv1}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambdaH2Inv2} H_{a_3} \text{NewScalar2}_{a_4} \text{T3HiggsPortal2CGInv2}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambda12Inv1} \text{NewScalar1}_{a_3} \text{NewScalar2}_{a_4} \text{T3CrossScalarCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambda12Inv2} \text{NewScalar1}_{a_3} \text{NewScalar2}_{a_4} \text{T3CrossScalarCGInv2}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{a_3} \text{NewScalar2}_{a_4} \text{T3Scalar2SelfCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar2}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar1}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{a_1}) \\
& + D_{\mu_1} (\text{NewScalar2}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{a_1})
\end{aligned}$$

$$\begin{aligned}
& \Delta \mathcal{L}_{\text{EFT}} - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{16 (M_F^2 - M_{\text{S1}}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{16 (M_F^2 - M_{\text{S2}}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{\text{S1}}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{2, i_3, r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{\text{S2}}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S2}}^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1}} \overline{\ell_{a_4, i_1}} {}^T C P_L \ell_{a_1, i_2} H_{a_2} H_{a_3} M_{\text{S2}}^2 \lambda_{\text{T3}} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_F^3}{4 (M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S1}}^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1}} \overline{\ell_{a_4, i_1}} {}^T C P_L \ell_{a_1, i_2} H_{a_2} H_{a_3} M_{\text{S1}}^2 \lambda_{\text{T3}} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_F^3}{4 (M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S2}}^2} \right) \overline{\lambda_{\text{T3}} \ell_{a_3, i_2}} C P_R \overline{\ell_{a_4, i_1}} {}^T M_{\text{S2}}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_2, a_4} \epsilon_{a_3, a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{4 (M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S1}}^2} \right) \overline{\lambda_{\text{T3}} \ell_{a_3, i_2}} C P_R \overline{\ell_{a_4, i_1}} {}^T M_{\text{S1}}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_2, a_4} \epsilon_{a_3, a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{4 (M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)}
\end{aligned}$$

$$\begin{aligned}
& + \frac{3h\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{8(M_F^2 - M_{S1}^2)} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{8(M_F^2 - M_{S1}^2)} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{16(M_F^2 - M_{S1}^2)^2} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{8(M_F^2 - M_{S1}^2)^2} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{8(M_F^2 - M_{S2}^2)} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{8(M_F^2 - M_{S2}^2)} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{16(M_F^2 - M_{S2}^2)^2} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}}{8(M_F^2 - M_{S2}^2)^2} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{8(M_F^2 - M_{S1}^2)} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{8(M_F^2 - M_{S1}^2)} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{16(M_F^2 - M_{S1}^2)^2} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{8(M_F^2 - M_{S1}^2)^2} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{8(M_F^2 - M_{S2}^2)} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{8(M_F^2 - M_{S2}^2)} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{16(M_F^2 - M_{S2}^2)^2} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{8(M_F^2 - M_{S2}^2)^2} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& - \frac{h \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{\ell_{a_4,i_1}}^T C P_L \ell_{a_1,i_2} H_{a_2} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{4(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)}
\end{aligned}$$

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$$\begin{aligned}
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_1}}^T C P_L \ell_{a_1,i_2} H_{a_2} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_2,a_4} \epsilon_{a_3,a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_2,a_4} \epsilon_{a_3,a_1} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{4 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12\epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{13 \hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{12\epsilon} \\
& - \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& + \frac{3 \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3 \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S1}^2)} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S1}^2)} \\
& - \frac{3 \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3 \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3 \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S2}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{\text{S}2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{8(M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{\text{S}2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16(M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{\text{S}2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16(M_F^2 - M_{\text{S}2}^2)^2} \\
& + C_{\text{H}2} H_{a_1} H_{a_1}^\dagger \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{\text{S}1}^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{\text{S}1}^2)} \\
& - \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{\text{S}1}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{\text{S}1}^2)^2} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{8(M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16(M_F^2 - M_{\text{S}2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{\text{lambdaH1Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{5\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& + \frac{\text{lambdaH2Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}
\end{aligned}$$

$$\begin{aligned}
& + \frac{5\text{lambdaH2Inv}2^2\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& + \frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{1}{2}\text{lambdaH1Inv1}^2\hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH1Inv2}^2\hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{1}{2}\text{lambdaH2Inv1}^2\hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH2Inv2}^2\hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S1}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S1}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& - \frac{\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S2}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S2}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2}
\end{aligned}$$

$$T3 - A, \quad \alpha = 0 : \quad S_1 = (1, 0), \quad S_2 = (3, 1), \quad F = (2, \tfrac{1}{2})$$

$$\begin{aligned}
& \textbf{BSM UV Lagrangian} - \text{NewScalar1NewScalar1}^\dagger M_{\text{S1}}^2 \\
& + \frac{1}{2}\text{lambdaS1Inv1NewScalar1}^2 \left(\text{NewScalar1}^\dagger\right)^2 \\
& + i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1}) \\
& - \overline{N_{a_1, r_1}} N_{a_1, r_1} M_F \\
& + \text{NewScalar1} \overline{\ell_{a_1, i_1}} C P_R \overline{N_{a_2, r_1}}^T y_{1, i_1, r_1} \epsilon_{a_1, a_2} \\
& + \overline{y_{2, i_1, r_1}} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1} \text{NewScalar2}_{A_1} \text{gen}(\text{SU2L(fund)})_{A_1, a_1, a_2} \\
& + \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{a_2, r_1} \bar{\epsilon}_{a_1, a_2} \text{NewScalar1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \text{lambdaH1Inv1NewScalar1} H_{a_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2} \text{NewScalar1}^\dagger H_{a_1}^\dagger \\
& + \overline{\lambda_{\text{T3}}} \text{NewScalar2}_{A_1} \epsilon_{a_3, a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3} \text{NewScalar1}^\dagger H_{a_1}^\dagger H_{a_2}^\dagger \\
& - M_{\text{S}2}^2 \text{NewScalar2}_{A_1} \text{NewScalar2}_{A_1}^\dagger \\
& + \overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1} y_{2, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \text{NewScalar2}_{A_1}^\dagger \\
& + \text{NewScalar1} H_{a_2} H_{a_3} \lambda_{\text{T3}} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3} \bar{\epsilon}_{a_1, a_2} \text{NewScalar2}_{A_1}^\dagger \\
& + \text{lambda12Inv1NewScalar1NewScalar2}_{A_2} \text{T3CrossScalarCGInv1}_{A_1, A_2} \text{NewScalar1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv1}_{A_1, A_3, A_2, A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv2NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv2}_{A_1, A_3, A_2, A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar1})^\dagger D_{\mu_1} (\text{NewScalar1}) \\
& + D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1}) \\
& \Delta \mathcal{L}_{\text{EFT}} - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{\text{S}1}^2)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{16 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{\text{S}1}^2)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{2, i_3, r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S}2}^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1}} \ell_{a_4, i_2}^T C P_L \ell_{a_2, i_1} H_{a_1} H_{a_3} M_{\text{S}2}^2 \lambda_{\text{T3}} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_F^3}{2 (M_F^2 - M_{\text{S}1}^2) (M_F^2 - M_{\text{S}2}^2) (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S}1}^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1}} \ell_{a_4, i_2}^T C P_L \ell_{a_2, i_1} H_{a_1} H_{a_3} M_{\text{S}1}^2 \lambda_{\text{T3}} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_F^3}{2 (M_F^2 - M_{\text{S}1}^2) (M_F^2 - M_{\text{S}2}^2) (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S}2}^2} \right) \overline{\lambda_{\text{T3}} \ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}^T M_{\text{S}2}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{\text{S}1}^2) (M_F^2 - M_{\text{S}2}^2) (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{\text{S}1}^2} \right) \overline{\lambda_{\text{T3}} \ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}^T M_{\text{S}1}^2 y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{\text{S}1}^2) (M_F^2 - M_{\text{S}2}^2) (M_{\text{S}1}^2 - M_{\text{S}2}^2)}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3 \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S2}^2)} \\
& + \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S2}^2)} \\
& + \frac{3 \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3 \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S2}^2)} \\
& + \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S2}^2)} \\
& + \frac{3 \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{16 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) B_{\mu_1\mu_2}^2 \\
& - \frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1} \right)^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \left(W_{\mu_1\mu_2}^{A_1} \right)^2 \\
& - \frac{1}{6} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) \left(W_{\mu_1\mu_2}^{A_1} \right)^2 \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S2}^2)} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S2}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_F^2 - M_{S_2}^2)^2} \\
& + \text{CH21} H_{a_1} H_{a_1}^\dagger \\
& + \hbar C_{\text{H2}} H_{a_2} \text{T3HiggsPortal1CGInv1}_{a_1,a_2} H_{a_1}^\dagger \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2 (M_F^2 - M_{S_1}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2 (M_F^2 - M_{S_1}^2)} \\
& - \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4 (M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{8 (M_F^2 - M_{S_2}^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{8 (M_F^2 - M_{S_2}^2)} \\
& - \frac{3\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3\text{lambdaH2Inv2}^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH2Inv2}^2 \hbar \log\left(\frac{\mu_{S_2}^{\text{bar}2}}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv1}^2 \hbar H_{a_3} H_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1,a_3} \text{T3HiggsPortal1CGInv1}_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \text{lambdaH1Inv1}^2 \hbar \log\left(\frac{\mu_{S_1}^{\text{bar}2}}{M_{S_1}^2}\right) H_{a_3} H_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1,a_3} \text{T3HiggsPortal1CGInv1}_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S1}^2 - M_{S2}^2)}
\end{aligned}$$

$$T3 - D, \quad \alpha = -2 : \quad S_1 = (3, -1), \quad S_2 = (1, 0), \quad F = (2, -\frac{1}{2})$$

$$\begin{aligned}
& \text{BSM UV Lagrangian} - \text{NewScalar1}_{A_1} \text{NewScalar1}_{A_1}^\dagger M_{S1}^2 \\
& + \frac{1}{2} \text{lambdaS2Inv1NewScalar2}^2 \left(\text{NewScalar2}^\dagger \right)^2 \\
& + \text{NewScalar2} \overline{y_{2,i_1,r_1}} \overline{N_{a_1,r_1}} P_L \ell_{a_1,i_1} \\
& + i \overline{N_{a_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1,r_1}) \\
& - \overline{N_{a_1,r_1}} N_{a_1,r_1} M_F \\
& + \overline{\ell_{a_3,i_1}} C P_R \overline{N_{a_1,r_1}}^T \text{NewScalar1}_{A_1} y_{1,i_1,r_1} \epsilon_{a_1,a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_3,a_2} \\
& - \text{NewScalar2} M_{S2}^2 \text{NewScalar2}^\dagger \\
& + \overline{\ell_{a_1,i_1}} P_R N_{a_1,r_1} y_{2,i_1,r_1} \text{NewScalar2}^\dagger \\
& + H_{a_2} H_{a_3} \text{NewScalar1}_{A_1} \lambda_{T3} \text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3} \bar{\epsilon}_{a_1,a_2} \text{NewScalar2}^\dagger \\
& + \text{lambdaH2Inv1NewScalar2} H_{a_2} T3 \text{HiggsPortal2CGInv1}_{a_1,a_2} \text{NewScalar2}^\dagger H_{a_1}^\dagger \\
& + \overline{y_{1,i_1,r_1}} \ell_{a_3,i_1}^T C P_L N_{a_1,r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_2,a_3} \bar{\epsilon}_{a_1,a_2} \text{NewScalar1}_{A_1}^\dagger \\
& + \text{lambda12Inv1NewScalar2NewScalar1}_{A_2} T3 \text{CrossScalarCGInv1}_{A_1,A_2} \text{NewScalar2}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{A_2} T3 \text{HiggsPortal1CGInv1}_{a_1,a_2,A_1,A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{A_2} T3 \text{HiggsPortal1CGInv1}_{a_1,a_2,A_2,A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} T3 \text{HiggsPortal1CGInv2}_{a_1,a_2,A_1,A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} T3 \text{HiggsPortal1CGInv2}_{a_1,a_2,A_2,A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \text{NewScalar2} \overline{\lambda_{T3}} \epsilon_{a_3,a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1NewScalar1}_{A_3} \text{NewScalar1}_{A_4} T3 \text{Scalar1SelfCGInv1}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv2NewScalar1}_{A_3} \text{NewScalar1}_{A_4} T3 \text{Scalar1SelfCGInv2}_{A_1,A_3,A_2,A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar2})^\dagger D_{\mu_1} (\text{NewScalar2}) \\
& + D_{\mu_1} (\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1})
\end{aligned}$$

$$\begin{aligned}
& \Delta \mathcal{L}_{\text{EFT}} - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{16 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{2 (M_F^2 - M_{S1}^2) (M_F^2 - M_{S2}^2) (M_{S1}^2 - M_{S2}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S2}^2)^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S2}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{16(M_F^2 - M_{S1}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4(M_F^2 - M_{S2}^2)^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{6}\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& +\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_F^2-M_{\text{S}1}^2\right)} \\
& -\frac{3\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_F^2-M_{\text{S}1}^2\right)} \\
& -\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_F^2-M_{\text{S}1}^2\right)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right)\overline{y_{1,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_F^2-M_{\text{S}1}^2\right)^2} \\
& +\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\
& -\frac{1}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\
& -\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{\text{S}2}^2\right)} \\
& -\frac{\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}2}^2y_{2,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{\text{S}2}^2\right)} \\
& -\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\left(M_F^2-M_{\text{S}2}^2\right)^2} \\
& -\frac{\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2}\right)\overline{y_{2,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{\text{S}2}^4y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\left(M_F^2-M_{\text{S}2}^2\right)^2} \\
& +C_{\text{H}2}H_{a_1}H_{a_1}^\dagger \\
& +\text{CH}21\hbar H_{a_2}\text{T3HiggsPortal2CGInv}1_{a_1,a_2}H_{a_1}^\dagger \\
& +\frac{3\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{3\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_{\text{S}1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{8\left(M_F^2-M_{\text{S}1}^2\right)} \\
& -\frac{3\hbar\log\left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_{\text{S}1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{8\left(M_F^2-M_{\text{S}1}^2\right)} \\
& -\frac{3\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_{\text{S}1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{16\left(M_F^2-M_{\text{S}1}^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{\text{S}1}^2)^2} \\
& + \frac{h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{\text{S}2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3 \text{lambdaH1Inv} 2^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv} 2^2 h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv} 1^2 h H_{a_3} H_{a_4} \text{T3HiggsPortal} 2 \text{CGInv} 1_{a_1,a_3} \text{T3HiggsPortal} 2 \text{CGInv} 1_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \text{lambdaH2Inv} 1^2 h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) H_{a_3} H_{a_4} \text{T3HiggsPortal} 2 \text{CGInv} 1_{a_1,a_3} \text{T3HiggsPortal} 2 \text{CGInv} 1_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{h \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{h \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& + \frac{h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}1}^2} \right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}1}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{h \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{h \log \left(\frac{\mu_{\text{bar}2}}{M_{\text{S}2}^2} \right) \overline{\lambda_{\text{T}3}} H_{a_1} H_{a_2} M_{\text{S}2}^2 \lambda_{\text{T}3} H_{a_1}^\dagger H_{a_2}^\dagger}{2 (M_{\text{S}1}^2 - M_{\text{S}2}^2)}
\end{aligned}$$

$$T3 - E, \quad \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (3, 1), \quad F = (2, \tfrac{1}{2})$$

$$\begin{aligned}
& \text{BSM UV Lagrangian} - \text{NewScalar} 1_{A_1} \text{NewScalar} 1_{A_1}^\dagger M_{\text{S}1}^2 \\
& + i \overline{N_{a_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1,r_1})
\end{aligned}$$

$$\begin{aligned}
& -\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F \\
& + \overline{y_{2, i_1, r_1}} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1} \text{NewScalar2}_{A_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \\
& + \overline{\ell_{a_3, i_1}} C P_R \overline{N_{a_1, r_1}}^T \text{NewScalar1}_{A_1} y_{1, i_1, r_1} \epsilon_{a_1, a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_3, a_2} \\
& + \overline{y_{1, i_1, r_1}} \ell_{a_3, i_1}^T C P_L N_{a_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_2, a_3} \bar{e}_{a_1, a_2} \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{A_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \overline{\lambda_{T3}} \text{NewScalar2}_{A_2} \epsilon_{a_3, a_2} \text{fStruct}(\text{SU2L})_{A_1, A_2, A_3} \text{gen}(\text{SU2L}(\text{fund}))_{A_3, a_1, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{A_3} \text{NewScalar1}_{A_4} \text{T3Scalar1SelfCGInv1}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{A_3} \text{NewScalar1}_{A_4} \text{T3Scalar1SelfCGInv2}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar1}_{A_2}^\dagger \\
& - M_{S_2}^2 \text{NewScalar2}_{A_1} \text{NewScalar2}_{A_1}^\dagger \\
& + \overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1} y_{2, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \text{NewScalar2}_{A_1}^\dagger \\
& - H_{a_2} H_{a_3} \text{NewScalar1}_{A_2} \lambda_{T3} \text{fStruct}(\text{SU2L})_{A_1, A_2, A_3} \text{gen}(\text{SU2L}(\text{fund}))_{A_3, a_1, a_3} \bar{e}_{a_1, a_2} \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, A_1, A_2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{A_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, A_2, A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger \\
& + \text{lambda12Inv1} \text{NewScalar1}_{A_3} \text{NewScalar2}_{A_4} \text{T3CrossScalarCGInv1}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + \text{lambda12Inv2} \text{NewScalar1}_{A_3} \text{NewScalar2}_{A_4} \text{T3CrossScalarCGInv2}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + \text{lambda12Inv3} \text{NewScalar1}_{A_3} \text{NewScalar2}_{A_4} \text{T3CrossScalarCGInv3}_{A_1, A_3, A_2, A_4} \text{NewScalar1}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv1}_{A_1, A_3, A_2, A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv2} \text{NewScalar2}_{A_3} \text{NewScalar2}_{A_4} \text{T3Scalar2SelfCGInv2}_{A_1, A_3, A_2, A_4} \text{NewScalar2}_{A_1}^\dagger \text{NewScalar2}_{A_2}^\dagger \\
& + D_{\mu_1} (\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1}) \\
& + D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1}) \\
& \Delta \mathcal{L}_{\text{EFT}} - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{16 (M_F^2 - M_{S_1}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{16 (M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^4}{16 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{i\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{i\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{i\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{i\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F^3}{2 (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)} \\
& + \frac{3\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8 (M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)} \\
& + \frac{3\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{16(M_F^2 - M_{S2}^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}{}^\dagger M_F^2}{8(M_F^2 - M_{S2}^2)^2} \\
& - \frac{i\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{i\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{i\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \bar{\ell}_{a_4,i_2}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{i\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \bar{\ell}_{a_4,i_2}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger M_F}{2(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{5\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{6\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& - \frac{1}{6} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& - \frac{1}{6} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& + \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{8(M_F^2 - M_{S1}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}1}^2} \right) \overline{y_{1,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}1}^2 y_{1,i2,r1} Y_{e,i3,i1}}{8 (M_F^2 - M_{\text{S}1}^2)} \\
& - \frac{3h \overline{y_{1,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}1}^4 y_{1,i2,r1} Y_{e,i3,i1}}{16 (M_F^2 - M_{\text{S}1}^2)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}1}^2} \right) \overline{y_{1,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}1}^4 y_{1,i2,r1} Y_{e,i3,i1}}{16 (M_F^2 - M_{\text{S}1}^2)^2} \\
& + \frac{3h \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} y_{2,i2,r1} Y_{e,i3,i1}}{16\epsilon} \\
& - \frac{3}{32} \overline{h y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} y_{2,i2,r1} Y_{e,i3,i1} \\
& - \frac{3h \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}2}^2 y_{2,i2,r1} Y_{e,i3,i1}}{8 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2} \right) \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}2}^2 y_{2,i2,r1} Y_{e,i3,i1}}{8 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3h \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}2}^4 y_{2,i2,r1} Y_{e,i3,i1}}{16 (M_F^2 - M_{\text{S}2}^2)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2} \right) \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{\text{S}2}^4 y_{2,i2,r1} Y_{e,i3,i1}}{16 (M_F^2 - M_{\text{S}2}^2)^2} \\
& + C_{\text{H}2} H_{a1} H_{a1}^\dagger \\
& + \frac{3h \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{1,i3,r1} H_{a1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \overline{h y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{1,i3,r1} H_{a1}^\dagger \\
& - \frac{3h \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{\text{S}1}^2 y_{1,i3,r1} H_{a1}^\dagger}{8 (M_F^2 - M_{\text{S}1}^2)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}1}^2} \right) \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{\text{S}1}^2 y_{1,i3,r1} H_{a1}^\dagger}{8 (M_F^2 - M_{\text{S}1}^2)} \\
& - \frac{3h \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{\text{S}1}^4 y_{1,i3,r1} H_{a1}^\dagger}{16 (M_F^2 - M_{\text{S}1}^2)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}1}^2} \right) \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{\text{S}1}^4 y_{1,i3,r1} H_{a1}^\dagger}{16 (M_F^2 - M_{\text{S}1}^2)^2} \\
& + \frac{3h \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{2,i3,r1} H_{a1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \overline{h y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{2,i3,r1} H_{a1}^\dagger \\
& - \frac{3h \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{\text{S}2}^2 y_{2,i3,r1} H_{a1}^\dagger}{8 (M_F^2 - M_{\text{S}2}^2)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_{\text{S}2}^2} \right) \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{\text{S}2}^2 y_{2,i3,r1} H_{a1}^\dagger}{8 (M_F^2 - M_{\text{S}2}^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3\text{lambdaH2Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv}2^2 \hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{3}{2} \text{lambdaH2Inv}2^2 \hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2}
\end{aligned}$$
